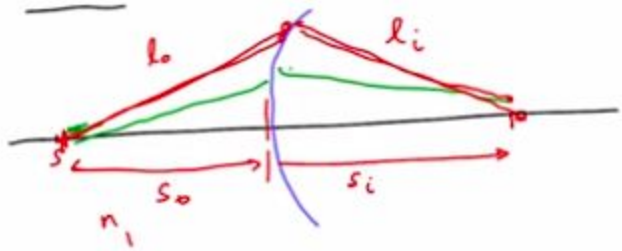
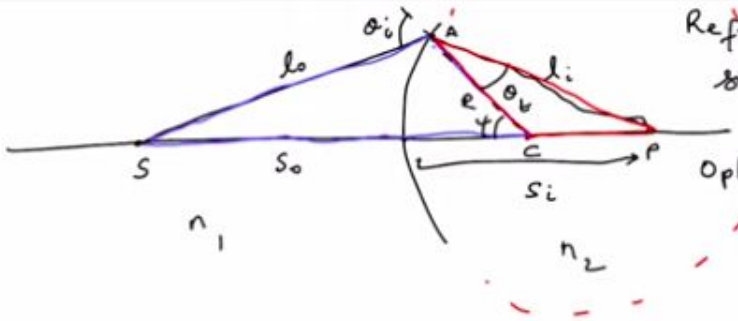
Focus

Cartesian Oval.



OPL $\boxed{n_1 s_o + n_2 s_i = n_1 l_o + n_2 l_i = \text{constant}}$

Most lenses → spherical.



Refraction through a single surface

Optical axis

$$OPL = n_1 l_o + n_2 l_i$$

$$= n_1 \left[R^2 + (s_o + R)^2 - 2R(s_o + R)\cos\psi\right]^{1/2} + n_2 \left[R^2 + (s_i - R)^2 + 2R(s_i - R)\cos\psi\right]^{1/2}$$

$$\frac{d(OPL)}{d\psi} = \frac{n_1}{2} \frac{+2R(s_o + R)\sin\psi}{l_o} - \frac{n_2}{2} \frac{2R(s_i - R)\sin\psi}{l_i}$$

$$= n_1 \left[R^2 + (S_o + R) - 2R(S_o+R)\cos\psi\right] + n_2 \ldots$$

$$\frac{d(OPL)}{d\psi} = \frac{n_1}{\cancel{2}} \frac{+\cancel{2}R(S_o+R)\cancel{\sin\psi}}{l_o} - \frac{n_2}{\cancel{2}} \frac{\cancel{2}R(S_i-R)\cancel{\sin\psi}}{l_i}$$

$$0 = \quad " \quad - \quad "$$

$$\frac{n_1}{l_o}(S_o+R) = \frac{n_2}{l_i}(S_i-R)$$

$$\frac{n_1}{l_o} + \frac{n_2}{l_i} = \frac{1}{R}\left[\frac{n_2 S_i}{l_i} - \frac{n_1 S_o}{l_o}\right]$$

$$0 = \quad " \quad - \quad "$$

$$\frac{n_1}{l_o}(S_o+R) = \frac{n_2}{l_i}(S_i-R)$$

$\cos\psi \approx 1$

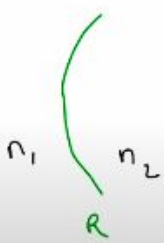
$l_o \approx \left[R^2 + (S_o+R)^2 - 2R(S_o+R)\right]^{1/2}$

$= \left[\cancel{R^2} + S_o^2 + \cancel{R^2} + 2RS_o - \cancel{2RS_o} - \cancel{2R^2}\right]^{1/2}$

$$\frac{n_1}{l_o} + \frac{n_2}{l_i} = \frac{1}{R}\left[\frac{n_2 S_i}{l_i} - \frac{n_1 S_o}{l_o}\right]$$

$l_o \approx S_o$

$l_i \approx S_i$



for the condition $\cos\psi \approx 1$

$$\boxed{\frac{n_1}{S_o} + \frac{n_2}{S_i} = \frac{1}{R}(n_2 - n_1)}$$

$$= \frac{1}{f} \quad (\text{Power} - \text{units } m^{-1})$$

Paraxial rays
First order approx

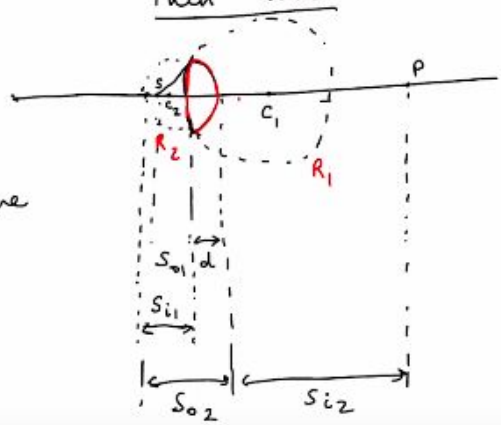
Thin Lens

S_o

f combined lens

f_1 would be the focal length if there was only one interface

$f < S_o < f_1$



Refraction at a single interface

$$\frac{n_2 - n_1}{R} = \frac{n_1}{S_o} + \frac{n_2}{S_i}$$

$S_o \downarrow \quad S_i \uparrow$

$S_o = f$

$S_i = \infty$

S_o decreases further?

$\boxed{S_i \rightarrow \text{negative}}$

focal length if there was only one interface

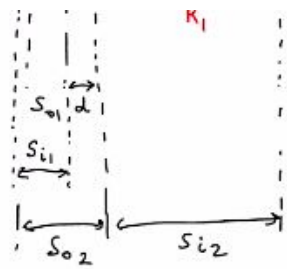
$f < S_o < f_1$

1st surface

$$\frac{n_m}{S_{o1}} + \frac{n_L}{S_{i1}} = \frac{n_L - n_m}{R_1}$$

2nd surface

$$\frac{n_L}{S_{o2}} + \frac{n_m}{S_{i2}} = \frac{n_m - n_L}{R_2}$$



$S_o = f$

$S_i = \infty$

S_o decreases further?

$\boxed{S_i \rightarrow \text{negative}}$

Sign Convention

	Lenses	Mirrors
Focal length (f)	+ Convex lens	+ Concave mirror
	- Concave lens	- Convex mirror
Object distance (d_o)	+ Object left of lens	+ Object in front of mirror
	- Object right of lens	- Object behind mirror (virtual object)
Image distance (d_i)	+ Image formed to the right of lens	+ Image in front of mirror
	- Image formed to the left of lens	- Image behind mirror (virtual object)
Magnification	+ Image upright with respect to object	+ Image upright with respect to object
	-Image inverted with respect to object	-Image inverted with respect to object

1st surface

$$\frac{n_m}{s_{o1}} + \frac{n_L}{s_{i1}} = \frac{n_L - n_m}{R_1} \quad (1)$$

2nd surface

$$\frac{n_L}{s_{o2}} + \frac{n_m}{s_{i2}} = \frac{n_m - n_L}{R_2} \quad (2)$$

s_{o2}

s_{o2} s_{i2}

$$|s_{o2}| = |s_{i1}| + d$$

$$s_{o2} = -s_{i1} + d$$

(1) + (2)

s_{i1} → negative

Sign Convention

obj dist to the left of the lens +ve

image dist to the left of the lens is −ve

$$n_m\left(\frac{1}{s_{o1}} + \frac{1}{s_{i2}}\right) = (n_L - n_m)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + n_L\left(\frac{-1}{s_{i1}} - \frac{1}{-s_{i1}+d}\right)$$

for a thin lens $(d \approx 0)$

$$n_m\left(\frac{1}{s_{o1}} + \frac{1}{s_{i2}}\right) = (n_L - n_m)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + n_L\left(\frac{-1}{s_{i1}} - \frac{1}{-s_{i1}+d}\right)$$

for a thin lens $(d \approx 0)$

medium $n_m = 1$

$$\frac{1}{s_{o1}} + \frac{1}{s_{i2}} = (n_L - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\boxed{\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \quad (\text{Power})}$$

u – obj dist.

v – image dist.